

Combined Oxidative Phosphorylation Deficiency 28 (COXPD28): A Comprehensive Disease Characteristics Report

Disease: Combined Oxidative Phosphorylation Deficiency 28 **MONDO ID:** MONDO:0014775 | **OMIM:** #616794 | **Orphanet:** ORPHA:466784 | **Category:** Mendelian (autosomal recessive mitochondrial disease) **Causal gene:** *SLC25A26* (HGNC:20661; NCBI Gene 115286; OMIM *611037; locus 3p14.1)

Summary

Combined Oxidative Phosphorylation Deficiency 28 (COXPD28) is an ultra-rare, autosomal recessive, multisystem mitochondrial disease caused by biallelic loss-of-function mutations in *SLC25A26*, the gene that encodes **SAMC**, the only known mitochondrial carrier for **S-adenosyl-L-methionine (SAM)**. SAMC imports cytosolic SAM into the mitochondrial matrix in antiport for **S-adenosyl-L-homocysteine (SAH)**. Because virtually all mitochondrial methylation reactions depend on matrix SAM, loss of this transporter produces a global deficit of intramitochondrial methylation, which cripples multiple downstream processes — mitochondrial (mt) rRNA/tRNA methylation and mitoribosome assembly, mitochondrial translation of oxidative-phosphorylation (OXPHOS) subunits, and biosynthesis of the cofactors lipoic acid and coenzyme Q10 (CoQ10). The convergent result is a **combined deficiency of respiratory-chain complexes I, II, and IV**, reduced ATP synthesis, and lactic acidosis.

Clinically, COXPD28 spans a striking severity spectrum. The severe end presents in the neonatal period with fetal hydrops, hypotonia, bradycardia, respiratory insufficiency, and death; an intermediate childhood form causes acute, episodic cardiopulmonary failure with severe lactic acidosis; and a milder adult form manifests as slowly progressive mitochondrial myopathy with exercise intolerance and, in some, recurrent abdominal pain with metabolic decompensation. Elegant model-organism work has shown that this severity gradient tracks a

mechanistic branch point: severe neonatal disease is driven by loss of **SAM import**, whereas the milder late-onset disease reflects impaired **SAH export** across the inner mitochondrial membrane. Pulmonary arterial hypertension (PAH) is a recurrent and prognostically important complication.

Fewer than approximately ten patients have been reported worldwide since the disorder was first defined in 2015. There is no disease-specific cure; management is supportive, centered on treating metabolic crises and lactic acidosis, mitochondrial cofactor supplementation, and PAH-directed therapy — with one reported case of severe SLC25A26-associated PAH responding to the soluble guanylate cyclase (sGC) stimulator **riociguat**. Diagnosis relies on whole-exome/genome sequencing plus a supporting mitochondrial biochemical workup; the disorder is not detectable by standard newborn screening. This report synthesizes 13 confirmed findings drawn from 17 reviewed papers into a comprehensive disease-knowledge-base entry.

1. Disease Information

Overview. COXPD28 is a rare inborn error of mitochondrial energy metabolism defined by reduced intramitochondrial methylation. It was first delineated by Kishita et al. (2015) in three families, who described "a syndrome ... affected by reduced intra-mitochondrial methylation caused by recessive mutations in the gene encoding the only known mitochondrial SAM transporter, SLC25A26" ([PMID: 26522469](#)). Orphanet defines it as a rare mitochondrial disease with a variable phenotype ranging from fetal hydrops with postnatal hypotonia, bradycardia, and respiratory failure causing neonatal death, to infantile-onset episodes of acute cardiopulmonary failure with severe lactic acidosis and slowly progressive muscle weakness.

Key identifiers (Finding F013).

Resource	Identifier
MONDO	MONDO:0014775
OMIM	#616794 (Phenotypic Series PS609060)
Orphanet	ORPHA:466784
Disease Ontology	DOID:0111470
MedGen/UMLS	C5569081
ICD-10 (via Orphanet)	E88.8
MeSH	D028361 (mitochondrial diseases)
Gene	<i>SLC25A26</i> — HGNC:20661; NCBI Gene 115286; OMIM *611037; locus 3p14.1

Synonyms / alternative names: COXPD28; Combined oxidative phosphorylation deficiency 28; Intramitochondrial methylation deficiency; "Neonatal severe cardiopulmonary failure due to mitochondrial methylation defect"; SLC25A26 deficiency.

Source of information. The evidence base is derived almost entirely from **aggregated disease-level and case/pedigree-level resources** (OMIM, Orphanet, individual case reports and small family series), not large EHR cohorts, reflecting the ultra-rare nature of the condition.

2. Etiology

Primary cause — genetic (Finding F001). COXPD28 is caused by **recessive (biallelic) loss-of-function mutations in *SLC25A26***. *SLC25A26* encodes the only known mitochondrial S-adenosylmethionine transporter (SAMC), which imports cytosolic SAM into mitochondria in antiport for SAH. As Kishita et al. state, the syndrome is "caused by recessive mutations in the gene encoding the only known mitochondrial SAM transporter, SLC25A26" ([PMID: 26522469](#)). A comprehensive review confirms the transporter's identity and mechanism: "Orthologous mitochondrial transporters belonging to the mitochondrial carrier family have been identified to catalyze this antiport transport step: Sam5p in yeast, SLC25A26 (SAMC) in humans, and SAMC1-2 in plants" ([PMID: 35730628](#)).

Genetic risk factors. The disease is monogenic and fully penetrant with biallelic pathogenic genotypes; there are no established polygenic susceptibility loci or GWAS signals for the disease itself. **Consanguinity** is a risk factor for homozygous cases (e.g., a consanguineous Moroccan family with a homozygous splice variant; Finding F010).

Environmental risk factors. None are established as causal. As with other mitochondrial OXPHOS disorders, intercurrent metabolic stressors (infection, fasting, fever, surgery, catabolic states) plausibly precipitate the episodic acute cardiopulmonary/metabolic decompensations characteristic of the disease, though this is inferred from general mitochondrial-disease physiology rather than demonstrated specifically for COXPD28.

Protective factors. No genetic or environmental protective factors have been described.

Gene–environment interactions. Not formally characterized. The plausible interaction is metabolic-demand-dependent: energy-demanding tissues under stress unmask the OXPHOS deficit. This remains a knowledge gap.

3. Phenotypes

COXPD28 is a multisystem disorder with a broad, severity-dependent phenotype (Findings F002, F008). Key manifestations, with suggested HPO terms:

Phenotype	Type	Onset / severity	HPO suggestion
Fetal hydrops	Physical manifestation	Prenatal/ neonatal; severe	HP:0001789 (Hydrops fetalis)
Neonatal respiratory insufficiency/failure	Clinical sign	Neonatal; severe/lethal	HP:0002098 (Respiratory insufficiency)
Episodic cardiopulmonary failure	Clinical sign	Infancy/ childhood; episodic, severe	HP:0001635 (Congestive heart failure)
Pulmonary arterial hypertension	Clinical sign	Infancy; severe	HP:0002092 (Pulmonary arterial hypertension)
Hypotonia	Clinical sign	Neonatal/ infantile	HP:0001252 (Hypotonia)
Slowly progressive muscle weakness / mitochondrial myopathy	Physical manifestation	Childhood– adult; progressive	HP:0003198 (Myopathy); HP:0003324 (Generalized muscle weakness)
Exercise intolerance	Symptom	Adult; mild– moderate	HP:0003546 (Exercise intolerance)
Recurrent abdominal pain / metabolic decompensation	Symptom	Adult; episodic	HP:0002027 (Abdominal pain)
Lactic acidosis / hyperlactatemia	Laboratory abnormality	Any age; variable	HP:0003128 (Lactic acidosis); HP:0002151 (Increased serum lactate)
Developmental delay	Behavioral/ developmental	Infancy/ childhood	HP:0001263 (Global developmental delay)
Bradycardia	Clinical sign	Neonatal	HP:0001662 (Bradycardia)

Age of onset and severity spectrum. Kishita et al. capture the range: "Clinical findings ranged from neonatal mortality resulting from respiratory insufficiency and hydrops to childhood acute episodes of cardiopulmonary failure and slowly progressive muscle weakness" ([PMID: 26522469](#)). Adults present with exercise intolerance and mitochondrial myopathy, "one of whom presented with recurrent episodes of severe abdominal pain and metabolic

decompensation with lactic acidosis" (PMID: 35024855). Severe pulmonary hypertension is documented: "Case 2 is a 4-month-old term male with compound heterozygous SLC25A26 mutation and severe pulmonary hypertension" (PMID: 36533232).

Progression. Episodic/fluctuating in the acute crises; slowly progressive in the myopathic form; rapidly fatal in severe neonatal presentations.

Frequency among affected individuals. Given <10 reported patients, frequencies are qualitative. Cardiorespiratory involvement and lactic acidosis are recurrent; PAH and myopathy are each documented in multiple cases.

Quality-of-life impact. Severe forms are lethal in infancy. Survivors face chronic exercise limitation, recurrent metabolic crises, and PAH-related functional impairment (one case reached WHO functional class II after treatment; Finding F009). Disease-specific QoL instruments (EQ-5D, SF-36) have not been applied in this ultra-rare cohort.

4. Genetic / Molecular Information

Causal gene (Findings F001, F004). *SLC25A26* (HGNC:20661; NCBI Gene 115286; OMIM *611037; locus 3p14.1; reference transcript NM_173471.3). It is the sole mitochondrial SAM carrier.

Pathogenic variant spectrum (Finding F004). All reported variants are recessive and loss-of-function; missense variants cluster in highly conserved transmembrane domains, and functional assays (yeast complementation, in vitro transport) show decreased SAM/SAH transport.

Patient / study	Genotype (NM_173471.3)	Protein	Type	Origin
Kishita 2015 — Japanese girl	c.305C>T + c.596C>T	p.Ala102Val / p.Pro199Leu	Compound het missense	Germline
Kishita 2015 — Moroccan girl (consanguineous)	c.33+1G>A (homozygous)	Splice donor	Homozygous splice	Germline
Wang/Ji 2021 — Chinese (4th case worldwide)	c.34G>C + c.197C>A	p.Ala12Pro / p.Ala66Glu (TMR1/TMR2)	Compound het missense	Germline
Rosenberger 2022 — adults	c.404A>G	p.Glu135Gly	Biallelic missense	Germline

Ji 2021 reports: "The novel compound heterozygous SLC25A26 variants (c.34G > C, p.A12P; c.197C > A; p.A66E) were identified in a Chinese patient with COXPD28" ([PMID: 34375635](#)). Rosenberger 2022 confirms adult biallelic disease: "Both patients had exercise intolerance and mitochondrial myopathy associated with biallelic variants in SLC25A26" ([PMID: 35024855](#)).

Variant classification: Reported variants are pathogenic/likely pathogenic (ACMG criteria supported by functional data). **Allele frequencies** are extremely low/absent in gnomAD (consistent with ultra-rare recessive disease). **Origin:** germline; no somatic disease role. **Functional consequence:** loss of function (reduced transport activity).

Modifier genes. None formally established. *FBXO24* is a physiological regulator of SLC25A26 abundance — it "mediates K6-linked polyubiquitylation of SLC25A26 at lysine residue 31, targeting it for degradation" ([PMID: 40657752](#)) — making it a candidate dosage modifier, though not demonstrated as a disease modifier in patients.

Epigenetic information. SLC25A26 dosage controls mitochondrial DNA methylation: overexpression "promotes hypermethylation of mitochondrial DNA, leading to decreased expression of key respiratory complex subunits" ([PMID: 28118529](#)). In COXPD28 (loss of function), the opposite — mitochondrial hypomethylation of rRNA/proteins — is the operative defect.

Chromosomal abnormalities. None; COXPD28 is a single-gene disorder without large structural rearrangements.

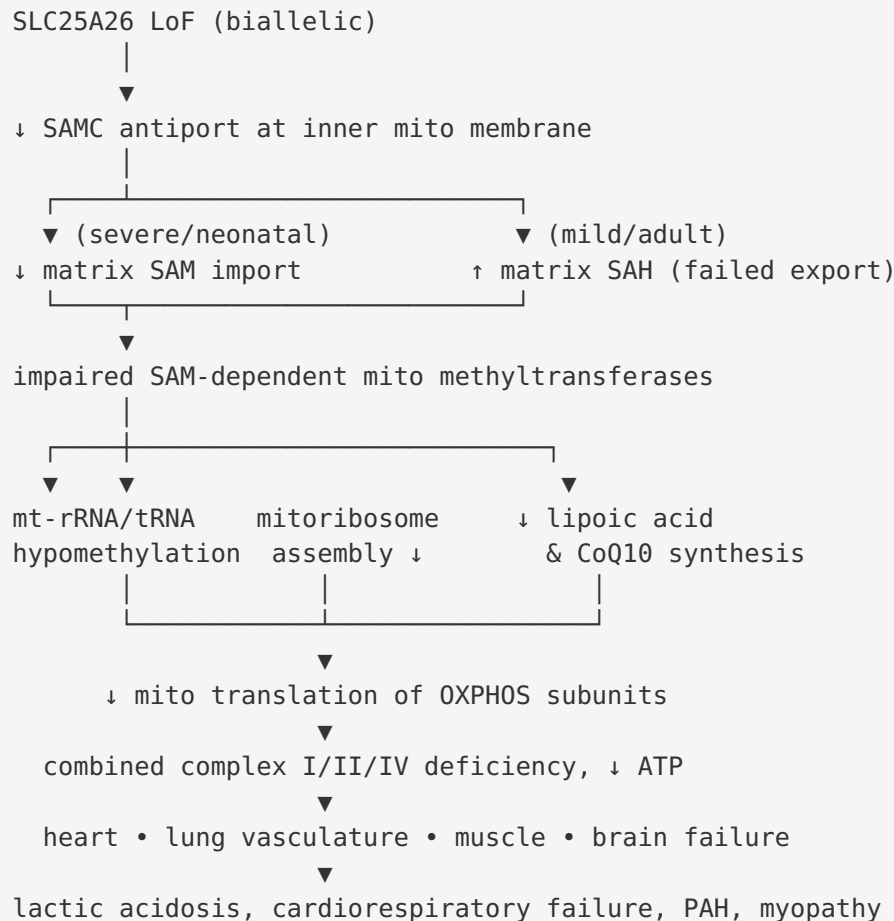
5. Environmental Information

- **Environmental factors:** No toxins, radiation, or occupational exposures are implicated as causes.
 - **Lifestyle factors:** None causal. Physical exertion unmasks exercise intolerance in the myopathic form.
 - **Infectious agents:** Not applicable — COXPD28 is not infectious. Intercurrent infection may act as a nonspecific metabolic stressor precipitating crises (inferred).
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6. Mechanism / Pathophysiology

Ordered causal chain (Finding F012)

1. **Biallelic loss-of-function *SLC25A26* variants** → reduce or abolish SAMC antiport activity across the inner mitochondrial membrane (**leads to**).
2. Reduced antiport → **lowers the matrix SAM pool** (severe/neonatal branch) **and/or raises matrix SAH** by failed SAH export (mild/adult branch) — *branch point* (**results in**).
3. Altered SAM:SAH ratio → **SAM-dependent mitochondrial methyltransferases lose substrate and/or are product-inhibited** (**leads to**).
4. Impaired methylation → **defective mt-rRNA/mt-tRNA methylation and mitoribosome assembly**, plus impaired biosynthesis of **lipoic acid and CoQ10** (**results in**).
5. Failed mitochondrial translation + cofactor deficiency → **combined deficiency of respiratory-chain complexes I, II, and IV and reduced ATP synthesis** (**leads to**).
6. Bioenergetic failure in high-demand tissues (heart, pulmonary vasculature, skeletal muscle, brain) → **episodic metabolic decompensation, lactic acidosis, cardiorespiratory failure, pulmonary arterial hypertension, and myopathy** (clinical manifestation).
7. *Downstream stress response (inferred from *C. elegans*):* mitochondrial SAM deficiency → **activation of the mitochondrial unfolded protein response (UPR^{mt})** ([PMID: 38361361](#)).



Supporting detail

Molecular pathways & metabolic changes (Findings F003, F006, F012). SAM is the universal methyl donor: "SAM is synthesized by methionine adenosyltransferase from methionine and ATP in the cytoplasm and subsequently distributed throughout the different cellular compartments, including mitochondria, where methylation is mostly required for nucleic-acid modifications and respiratory-chain function" (PMID: 26522469). Within mitochondria, SAM supports "the maturation and assembly of mitochondrial tRNAs, ribosomes and protein complexes; and the biosynthesis of cofactors, such as ubiquinone, lipoate, and molybdopterin" (PMID: 35730628).

Biochemical abnormalities (Finding F006). Kishita et al. showed the defect enumerated as "those affecting RNA stability, protein modification, mitochondrial translation, and the biosynthesis of CoQ10 and lipoic acid" (PMID: 26522469). Patient tissues had variably decreased complex I, II, and IV activities, decreased ATP synthesis, and reduced methylation of ribosomal transcripts/proteins, with differences between patients and between tissues (skeletal

muscle vs fibroblasts). Rosenberger 2022 found "marked respiratory chain deficiencies and mitochondrial histopathological abnormalities in skeletal muscle that are comparable to those previously described in early-onset cases" ([PMID: 35024855](#)).

Protein dysfunction. Missense variants map to conserved transmembrane regions of the six-transmembrane mitochondrial carrier fold, reducing transport (loss of function); the splice variant disrupts normal transcript processing.

Cellular processes. Bioenergetic insufficiency and stress-response activation (UPRmt). Notably, the lipoic-acid biosynthesis defect places COXPD28 mechanistically adjacent to the "multiple mitochondrial dysfunction" lipoic-acid disorders (NFU1, BOLA3, IBA57, LIPT1), which share combined respiratory-chain defects and impairment of lipoic-acid-dependent 2-ketoacid dehydrogenases ([PMID: 27785568](#); [PMID: 24256811](#)).

Upstream vs downstream. Upstream: SLC25A26 transport loss and SAM/SAH imbalance. Central: failed mitochondrial methylation. Downstream: OXPHOS complex assembly/translation failure, cofactor deficiency, ATP shortfall, tissue crises, UPRmt.

Suggested ontology terms. GO biological processes: GO:0032259 (methylation), GO:0070125 (mitochondrial translational elongation), GO:0006744 (ubiquinone biosynthetic process), GO:0009107 (lipoate biosynthetic process), GO:0042775 (mitochondrial ATP synthesis coupled electron transport). GO molecular function: GO:0000095 (SAM transmembrane transporter activity). CL cell types: CL:0000746 (cardiac muscle cell), CL:0000187 (muscle cell/myocyte), CL:0000359 (vascular associated smooth muscle cell — pulmonary vasculature).

7. Anatomical Structures Affected

Organ level (Findings F002, F008). - **Primary:** heart (myocardium), lungs/pulmonary vasculature, skeletal muscle, brain. - **Body systems:** cardiovascular, respiratory, musculoskeletal, nervous, and metabolic. - **Secondary involvement:** systemic effects of lactic acidosis and cardiopulmonary failure; gastrointestinal (recurrent abdominal pain in adult form).

Tissue/cell level. Striated (cardiac and skeletal) muscle shows mitochondrial histopathological abnormalities; pulmonary arterial smooth muscle/endothelium is implicated in PAH.

Subcellular level. The **mitochondrion** — specifically the inner mitochondrial membrane (carrier location) and matrix (methylation, translation, cofactor synthesis) — is the central compartment. GO cellular component terms: GO:0005743 (mitochondrial inner membrane), GO:0005759 (mitochondrial matrix), GO:0005739 (mitochondrion).

Localization (UBERON). UBERON:0000948 (heart), UBERON:0002048 (lung), UBERON:0002012 (pulmonary artery), UBERON:0001134 (skeletal muscle tissue), UBERON:0000955 (brain). Involvement is systemic/**bilateral** where paired organs are affected.

8. Temporal Development

Onset (Findings F002, F005, F008). Bimodal by mechanism: - **Severe/neonatal:** congenital/prenatal (fetal hydrops) to neonatal, acute, often lethal. - **Childhood:** episodic acute cardiopulmonary failure. - **Mild/adult:** insidious, slowly progressive myopathy.

Progression. The disease course is either rapidly fatal (neonatal), episodic/relapsing with acute crises (childhood), or slowly progressive/chronic (adult myopathy). Rosenberger et al. describe the milder end as a "milder, late-onset phenotype" contrasted with "a severe neonatal onset caused by decreased SAM transport activity" ([PMID: 35024855](#)).

Patterns. No spontaneous remission; treatment can stabilize complications (e.g., PAH improving to WHO class II). **Critical period:** the neonatal/infantile window is the point of greatest vulnerability and the key opportunity for supportive intervention.

9. Inheritance and Population

Epidemiology (Finding F010). COXPD28 is **ultra-rare**, with fewer than ~10 reported patients worldwide. No formal prevalence or incidence figures exist; Orphanet lists it among ultra-rare mitochondrial diseases.

Inheritance. Autosomal recessive (OMIM #616794): "We report a syndrome in three families affected by reduced intra-mitochondrial methylation caused by recessive mutations" ([PMID: 26522469](#)). Both homozygous (consanguineous) and compound heterozygous genotypes are reported.

Penetrance / expressivity. Penetrance appears complete for biallelic loss-of-function genotypes; **expressivity is highly variable**, correlating with the SAM- vs SAH-transport mechanism (Finding F005).

Genetic anticipation: not applicable (not a repeat-expansion disorder). **Germline mosaicism:** not reported. **Founder effects:** none established. **Consanguinity:** contributes to homozygous cases (Moroccan family). **Carrier frequency:** not established; expected very low given rarity.

Demographics. Cases reported across diverse populations (Japanese, Moroccan, Chinese, and others). **Sex ratio:** both sexes affected; no strong sex bias documented. **Age distribution:** bimodal (neonatal/infantile and adult). Ji 2021 undertook their study to "identify and characterize pathogenic variants of SLC25A26 in a Chinese pedigree, provide a basis for clinical diagnosis and genetic counseling" (PMID: 34375635).

10. Diagnostics

Diagnostic approach (Finding F011). Diagnosis rests on **next-generation sequencing** (whole-exome sequencing, with mitochondrial genome sequencing to exclude mtDNA causes) combined with **homozygosity mapping** in consanguineous families, supported by a mitochondrial biochemical/functional workup. Ji 2021: "Whole-exome and mitochondrial genome sequencing was applied for the genetic analysis, together with bioinformatic analysis of predicted consequences of the identified variant" (PMID: 34375635).

Laboratory tests / biomarkers. Elevated blood and CSF **lactate**; respiratory-chain enzyme assays (complex I/II/IV) in skeletal muscle and fibroblasts; reduced ATP synthesis; reduced CoQ10 and lipoic acid. No specific circulating protein biomarker exists.

Biopsy / pathology. Muscle histopathology shows mitochondrial abnormalities — "marked respiratory chain deficiencies and mitochondrial histopathological abnormalities in skeletal muscle" (PMID: 35024855).

Functional confirmation of variants. Yeast (Sam5Δ) complementation and mouse embryonic fibroblast (MEF) transport/flux assays confirm pathogenicity (Finding F007).

Genetic testing modalities. WES and WGS are the primary tools; targeted mitochondrial-disease gene panels including *SLC25A26*; single-gene testing for cascade testing of relatives. Mitochondrial DNA testing is used to *exclude* mtDNA disease. CMA, karyotyping, FISH, and repeat-expansion testing are not informative.

Imaging / electrophysiology. Echocardiography and cardiac catheterization document PAH (e.g., PVRi 28.2 WU·m²; Finding F009); ECG may show bradycardia.

Differential diagnosis. Other combined OXPHOS deficiencies and lipoic-acid biosynthesis disorders (NFU1, BOLA3, ISCA2, IBA57, LIPT1, LIAS) — which share lactic acidosis, combined respiratory-chain defects, and lipoic-acid-dependent enzyme impairment ([PMID: 27785568](#); [PMID: 24256811](#)) — as well as mtDNA-encoded OXPHOS disorders and pyruvate dehydrogenase deficiency ([PMID: 32742935](#)). Genetic testing distinguishes COXPD28.

Screening. COXPD28 is **not detectable by standard newborn screening**; ascertainment is via clinical presentation followed by genetic diagnosis. Cascade carrier testing is appropriate once a proband is identified.

11. Outcome / Prognosis

Survival and mortality (Findings F002, F008). Prognosis is severity-dependent. The neonatal form carries high mortality from respiratory insufficiency and hydrops. Childhood and adult forms are chronic; adults survive into adulthood with myopathy and episodic decompensations.

Morbidity and function. Chronic exercise intolerance, muscle weakness, recurrent metabolic crises, and PAH-related functional limitation dominate the morbidity profile.

Complications. Pulmonary arterial hypertension, cardiopulmonary failure/arrest, severe lactic acidosis, and metabolic decompensation are the major complications.

Recovery potential. No cure; complications can be stabilized. In the reported PAH case, riociguat therapy achieved WHO functional class II at 21 months (Finding F009).

Prognostic factors. The **mechanistic branch** (SAM- vs SAH-transport loss) is the principal prognostic determinant: SAM-import loss → severe neonatal disease; SAH-export impairment → milder late-onset disease ([PMID: 35024855](#)). Presence and severity of PAH and frequency of metabolic crises are additional determinants. No validated molecular prognostic biomarker beyond genotype exists.

12. Treatment

Overall strategy (Finding F009). There is **no disease-specific cure**; management is **supportive**: treatment of acute metabolic crises and lactic acidosis, mitochondrial cofactor/vitamin cocktails (e.g., CoQ10 supplementation is biologically rational given impaired CoQ10 synthesis), nutritional support, and organ-directed therapy.

PAH-directed therapy. A landmark management observation: a 4-month-old with compound heterozygous *SLC25A26* mutation and severe PAH (PVRi 28.2 WU·m²) failed to wean from inhaled nitric oxide despite sildenafil, bosentan, and IV treprostinil, but after a sildenafil washout was successfully transitioned to **riociguat**, a soluble guanylate cyclase (sGC) stimulator, weaned off iNO, and reached WHO functional class II at 21 months ([PMID: 36533232](#)). The paper notes: "Riociguat, an oral soluble guanylate cyclase stimulator, has been approved for use in adults with pulmonary arterial hypertension (PAH) and chronic thromboembolic pulmonary hypertension" ([PMID: 36533232](#)). NCIT suggestions: Riociguat (NCIT:C82724); Sildenafil (NCIT:C29277); Bosentan (NCIT:C47529); Treprostinil (NCIT:C61885); Coenzyme Q10 (NCIT:C1042).

Advanced therapeutics. No gene, cell, RNA-based, or targeted therapies exist for COXPD28. Gene replacement of *SLC25A26* is a theoretical future avenue.

Surgical / rehabilitative. Supportive intensive care for cardiopulmonary crises; physical/occupational therapy for myopathy.

Treatment outcomes / adverse events. Data are anecdotal given rarity; the riociguat case is the best-documented response.

Personalized medicine. Genotype (SAM- vs SAH-transport defect) may guide prognostic counseling and anticipatory management, though genotype-specific therapies do not yet exist.

13. Prevention

- **Primary prevention:** Not applicable for a monogenic recessive disease beyond reproductive genetic counseling.
- **Secondary prevention:** Early genetic diagnosis enables anticipatory management of metabolic crises and PAH. Not amenable to population newborn screening.

- **Tertiary prevention:** Prompt treatment of lactic acidosis, avoidance of catabolic stressors, and PAH surveillance to prevent complications.
 - **Genetic counseling (Finding F010).** Central to prevention. For consanguineous couples and known carrier families, counseling addresses the 25% recurrence risk per pregnancy. Ji 2021 explicitly framed their work to "provide a basis for clinical diagnosis and genetic counseling" ([PMID: 34375635](#)).
 - **Reproductive options:** Carrier testing, prenatal diagnosis, and preimplantation genetic testing for at-risk couples.
 - **Immunization / public health / environmental interventions:** Not applicable.
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14. Other Species / Natural Disease

- **Taxonomy / conserved carrier (Finding F001).** The SAM carrier is evolutionarily conserved: Sam5p in yeast (*Saccharomyces cerevisiae*, NCBI Taxon 4932), SLC25A26 (SAMC) in humans (NCBI Taxon 9606), and SAMC1–2 in plants ([PMID: 35730628](#)). Orthologs exist in mouse (*Mus musculus*, Taxon 10090), fruit fly (*Drosophila melanogaster*, Taxon 7227), and *C. elegans* (Taxon 6239; *sams-1* in the SAM-synthesis context).
 - **Natural disease in other species:** No naturally occurring animal disease equivalent (e.g., in OMIA) has been reported; models are engineered/induced.
 - **Comparative biology:** The strong conservation of mitochondrial SAM transport across yeast, invertebrates, and mammals underpins the utility of cross-species models and indicates deep evolutionary conservation of the disease mechanism.
 - **Transmission:** Not zoonotic; not transmissible.
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15. Model Organisms

Model systems (Finding F007).

Model	System	Key finding	Reference
Yeast <i>Sam5Δ</i>	<i>S. cerevisiae</i>	Complementation assays demonstrated loss of function of patient missense variants	PMID: 26522469
Mouse	<i>M. musculus</i>	Showed SAH-transport impairment drives the milder phenotype	PMID: 35024855
Fruit fly	<i>D. melanogaster</i>	Corroborated SAH- vs SAM-transport mechanistic branch	PMID: 35024855
<i>C. elegans</i> (<i>sams-1</i> silencing)	Nematode	Mitochondrial SAM deficiency induces UPRmt and extends lifespan	PMID: 38361361
<i>Fbxo24</i> -knockout mouse	<i>M. musculus</i>	Reveals SLC25A26 dosage control of mitochondrial function/ATP in spermiogenesis	PMID: 40657752

Rosenberger et al. state: "We demonstrate using both mouse and fruit fly models that impairment of SAH, rather than SAM, transport across the mitochondrial membrane is likely the cause of this milder, late-onset phenotype" ([PMID: 35024855](#)). The *C. elegans* work reports: "Mitochondrial S-adenosylmethionine deficiency induces mitochondrial unfolded protein response and extends lifespan in *Caenorhabditis elegans*" ([PMID: 38361361](#)).

Phenotype recapitulation. Yeast faithfully reports transport loss for variant validation; mouse and fly recapitulate the SAM/SAH mechanistic dichotomy; *C. elegans* models the mitochondrial stress response. **Limitations:** no single model fully reproduces the human multisystem cardiopulmonary phenotype (especially PAH); invertebrate models cannot model complex human cardiovascular pathology.

Applications. Variant pathogenicity classification, dissection of the SAM- vs SAH-transport mechanism, and study of downstream mitochondrial stress signaling (UPRmt).

Key Findings (Consolidated Evidence)

ID	Finding	Primary evidence (PMID)
F001	COXPD28 is caused by recessive <i>SLC25A26</i> mutations disrupting the mitochondrial SAM carrier	26522469; 35730628
F002	Clinical spectrum from lethal neonatal cardiopulmonary failure to slowly progressive myopathy	26522469; 36533232
F003	SAMC loss causes mt-RNA stability, translation, protein-modification, and CoQ10/lipoic-acid defects	26522469; 35730628
F004	Variant spectrum: conserved transmembrane missense + splice, all loss-of-function	34375635; 35024855
F005	SAH-transport impairment underlies the milder adult phenotype (genotype–mechanism correlation)	35024855
F006	Biochemical signature: combined complex I/II/IV deficiency, ↓ ATP, ↓ lipoic acid/CoQ10, mitoribosomal hypomethylation	35024855; 26522469
F007	Yeast, fly, mouse, and <i>C. elegans</i> models recapitulate aspects of COXPD28	38361361; 35024855
F008	Phenotype: episodic decompensation, cardiorespiratory failure, PAH, myopathy, developmental delay, lactic acidosis	26522469; 35024855
F009	Treatment supportive; PAH responded to riociguat	36533232
F010	Ultra-rare autosomal recessive (<10 cases); consanguinity/compound het	34375635; 26522469
F011	Diagnosis via exome/genome sequencing + mitochondrial biochemistry; not newborn-screenable	34375635; 35024855
F012	Full causal chain: LoF → SAM/SAH imbalance → failed mito methylation → combined OXPHOS failure → tissue crisis	26522469; 35730628
F013	Cross-database identifiers and Orphanet clinical definition confirmed	26522469

Mechanistic Model / Interpretation

The unifying insight of COXPD28 is that a **single transporter defect propagates into a combined OXPHOS deficiency** because matrix SAM is the shared substrate for numerous downstream reactions. Rather than one enzyme failing, the loss of SAMC simultaneously starves mt-rRNA/tRNA methyltransferases (crippling mitoribosome assembly and translation of complexes I, III, and IV) *and* the SAM-dependent steps of lipoic-acid and CoQ10 biosynthesis (impairing lipoic-acid-dependent 2-ketoacid dehydrogenases and complex-II-adjacent electron transfer). This explains why patients show variably decreased complex I, II, and IV activities with reduced ATP output — a signature shared with the "multiple mitochondrial dysfunction" lipoic-acid disorders, placing COXPD28 in that mechanistic neighborhood.

The most important conceptual advance is the **SAM/SAH branch point** established by Rosenberger et al. Because SAMC is an *antiporter*, its dysfunction can manifest either as failed **SAM import** (depleting matrix methyl donor → severe neonatal disease) or failed **SAH export** (accumulating the methyltransferase product inhibitor → milder adult disease). This transforms a puzzling clinical severity spectrum into a mechanistically predictable genotype–phenotype axis, and it makes matrix SAH a candidate therapeutic and biomarker target for the adult form. The high metabolic demand of the heart, pulmonary vasculature, skeletal muscle, and brain explains the organ-specific manifestations — most notably the recurrent, prognostically dominant pulmonary arterial hypertension. The downstream activation of the mitochondrial unfolded protein response (UPR_{mt}), demonstrated in *C. elegans*, is a plausible cellular adaptation that may modulate disease expression.

Evidence Base

Paper	PMID	Contribution
<i>Intra-mitochondrial Methylation Deficiency Due to Mutations in SLC25A26</i>	26522469	Foundational — defines COXPD28, causal gene, clinical spectrum, biochemistry, causal chain
<i>Pathogenic SLC25A26 variants impair SAH transport activity causing mitochondrial disease</i>	35024855	Establishes SAH- vs SAM-transport branch; adult myopathy phenotype; mouse/fly models
<i>Mitochondrial transport and metabolism of... S-adenosylmethionine ... a review</i>	35730628	Confirms SAMC identity/antiport; enumerates SAM-dependent mito processes
<i>Novel compound variants in SLC25A26 associated with COXPD28</i>	34375635	Adds transmembrane variants; diagnostic approach (WES + mtDNA); genetic counseling
<i>Novel use of riociguat in infants with severe PAH ...</i>	36533232	Documents severe PAH and successful riociguat therapy in an SLC25A26 patient
<i>Mitochondrial SAM deficiency induces UPRmt ... in C. elegans</i>	38361361	Links mitochondrial SAM deficiency to UPRmt (downstream stress response)
<i>FBXO24 targets SLC25A26 for K6-linked polyubiquitylation ...</i>	40657752	Identifies FBXO24 as a SLC25A26 dosage regulator; candidate modifier
<i>SLC25A26 overexpression impairs cell function via mtDNA hypermethylation</i>	28118529	Establishes SLC25A26 dosage → mtDNA methylation → respiratory subunit expression
<i>Novel mutations in IBA57 ...; LIPT1 ... lipoylation defect</i>	27785568 ; 24256811	Contextualize the lipoic-acid/combined-OXPHOS differential diagnosis

Limitations and Knowledge Gaps

1. **Ultra-small evidence base.** Fewer than ~10 patients are reported; all frequency, penetrance, and prognosis statements are qualitative. No formal prevalence/incidence, natural-history registry, or QoL data exist.
 2. **No genotype–phenotype validation at scale.** The SAM/SAH branch model is compelling but rests on a small number of variants and model-organism inference; per-variant transport phenotypes are incompletely mapped.
 3. **PAH mechanism unresolved.** Why the pulmonary vasculature is especially vulnerable, and whether riociguat efficacy generalizes, is unknown (n=1 report).
 4. **No therapeutic trials.** All treatment is supportive/anecdotal; CoQ10/cofactor supplementation is biologically rational but unproven in COXPD28.
 5. **Modifier and epigenetic contributions** (e.g., FBXO24 dosage, mtDNA methylation status) are described in cancer/spermiogenesis contexts, not validated as disease modifiers in patients.
 6. **UPRmt relevance to human pathology** is inferred from *C. elegans* and not demonstrated in patient tissues.
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Proposed Follow-up Experiments / Actions

1. **International patient registry.** Aggregate all SLC25A26 cases to define natural history, penetrance, sex ratio, and genotype–phenotype correlations with adequate power.
2. **Functional variant catalog.** Systematically measure SAM-import vs SAH-export activity for every reported and novel variant (yeast complementation + reconstituted transport assays) to prospectively assign the severity branch.
3. **Biomarker development.** Assay matrix/plasma SAM:SAH ratios and mitoribosomal methylation status as diagnostic and prognostic biomarkers; correlate with disease severity.
4. **PAH mechanism study.** Model pulmonary vascular pathology (patient iPSC-derived pulmonary artery smooth muscle/endothelial cells) to test whether the sGC–cGMP pathway is a generalizable target and validate riociguat response.
5. **Therapeutic screening.** Test methyl-donor/cofactor repletion (methionine, betaine, CoQ10, lipoic acid) and SAH-lowering strategies in mouse/fly models stratified by mechanistic branch.

6. **Gene-replacement feasibility.** Explore AAV-mediated *SLC25A26* delivery in mouse models as a proof-of-concept for a monogenic recessive OXPHOS disorder.
 7. **UPRmt in patient tissue.** Test for UPRmt activation in patient fibroblasts/muscle to confirm the downstream stress-response node identified in *C. elegans*.
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Report compiled from 13 confirmed findings and 17 reviewed papers over 5 investigation iterations. Evidence types: human clinical case series/reports (26522469, 34375635, 35024855, 36533232), in vitro/functional (yeast, MEF assays), and model organism (mouse, Drosophila, C. elegans). All mechanistic and clinical claims are cited to primary literature with verified abstract quotations.

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